STRUCTURAL ASSESSMENT OF STEEL PROCESS EQUIPMENT USING FINITE ELEMENT ANALYSIS AND FATIGUE EVALUATION

Lappeenranta–Lahti University of Technology LUT

Bachelor’s Double Degree Programme in Mechanical Engineering, Bachelor's thesis

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ABSTRACT

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Structural assessment of steel process equipment using finite element analysis and fatigue evaluation

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As global competition intensifies and environmental sustainability becomes increasingly important, equipment manufacturers must find ways to optimize their products to improve material efficiency and minimize costs. This thesis focuses on equipment used in the process industry and explores opportunities for structural optimization using finite element analysis (FEA).

Initial assessment of the equipment suggests that current designs may be overly conservative. The thesis aims to validate this assessment using analytical calculations and static simulations. Subsequently, further simulations are conducted on a refined structure to determine the optimal material usage while maintaining an acceptable factor of safety. This is done by analysing stress distributions and deformation behaviour of the simulation models.

The findings suggest that there is potential for optimization in current designs. Although the study is limited to a static analysis, the results provide enough justification for equipment manufacturers to re-consider existing designs, thus promoting more efficient and sustainable design practices.

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DECLARATIONS

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1 Introduction

The design of industrial equipment plays a key role in ensuring safe, efficient and cost-effective operations in various manufacturing industries. As customer requirements and competition continue to increase, there is a need for material-efficient structural designs that comply with industry safety standards. This thesis focuses on equipment used in the process industry.

1.1 Research Background

Equipment used in the processing industries relies extensively on stainless steel as a robust material due to its excellent corrosion and durability properties (Kobrin, 1998, 4). It is well suited for handling materials with diverse chemical compositions and for operations requiring sterile conditions and high temperatures that would not be achievable using other materials. Furthermore, the favourable strength to weight ratio of steel is advantageous for industrial equipment, as it facilitates easier transportation, installation, and maintenance (Abambres and Arruda 2016).

Advancements in steel equipment manufacturing methods and material alloys have made it possible to develop lightweight yet effective designs. Modern design approaches have moved beyond the traditional code-based methods and instead emphasize performance-based design tailored to the specific function of the equipment. By minimizing overdesign, engineers can achieve acceptable factors of safety using less material, thereby leading to lower overall manufacturing cost (Abambres and Arruda 2016).

This is where the use of commercially available finite element analysis (FEA) software becomes essential. Computer Aided Design (CAD) tools and simulation software provide a less time consuming and cheaper alternative to performing laboratory experiments which may not always be feasible for equipment manufacturers (Abambres and Arruda 2016).

1.2 Research Problem

Current industrial equipment used in the process industry is often designed in accordance with the Eurocodes. While this may ensure reliability, it can also lead to overdesigned structures and excessive material usage. This results in high manufacturing cost and equipment weight, which impact its maintenance and transport. In an increasingly competitive landscape, where equipment is becoming more compact and environmental concerns are intensifying, it is vital that components are designed using an optimal amount of material to fulfil their intended function, enabling companies to remain competitive in the marketplace.

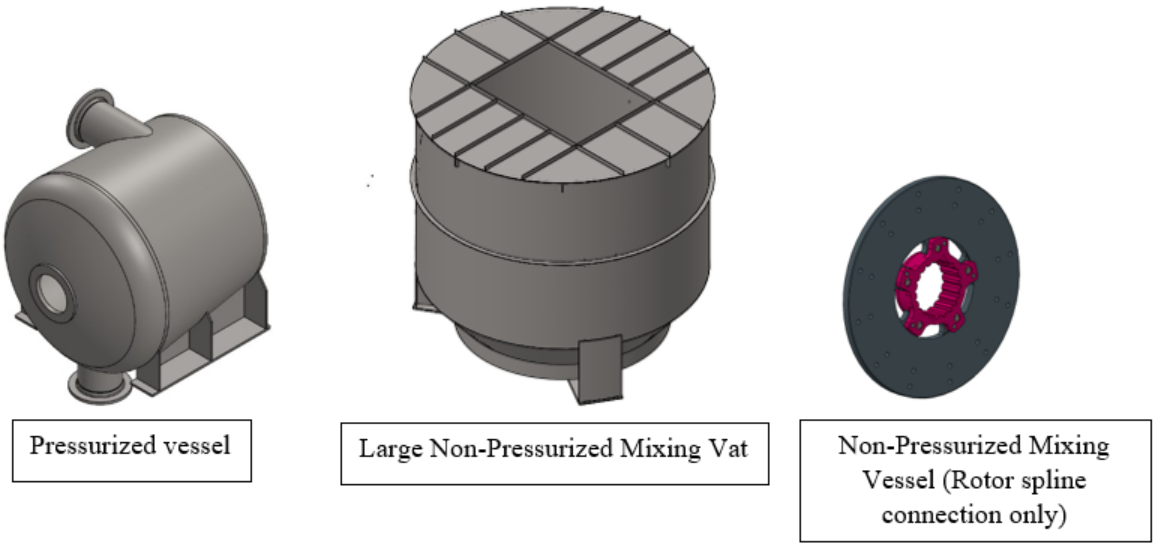


Figure 1 - Equipment under analysis

This thesis will focus on the three types of equipment shown in Figure 1. These choices represent common equipment types used in the process industries. Therefore, their analyses offer insights into the structural design of other equipment that operate under similar conditions.

1.3 Research Questions

1. Is the current equipment overdesigned?
2. In which areas can material usage be optimized while maintaining current capacity and acceptable safety factors?

3. For the non-pressurized mixing vessel, what are the characteristics of the cyclic loading that could lead to fatigue failure of the mechanical connection within a specified service life?

1.4 Scope and Limitations

This study focuses on the structural optimization of selected process equipment. The analysis includes analytical calculations and simulations using the finite element method.

The scope of the analysis for both the pressurized vessel and non-pressurized mixing vat is limited to structural behaviour under static loading. This assumes a linear elastic model and does not consider material defects, residual stresses or plastic deformation. Additionally, some geometrical features such as bolt holes and welds were omitted to simplify the simulations. Finally, only loads that could be represented as static were considered, therefore any cyclic loading or corrosion effects are not included.

The fatigue assessment is based solely on analytical calculations, providing an approximate evaluation of the cyclic loading. More detailed analysis would be required to obtain a more accurate description of the fatigue behaviour.

Therefore, the results obtained in this study depict a sufficiently accurate representation of the real-world conditions. However, they should be interpreted in light of the stated limitations.

1.5 Expected Outcomes and Significance

The study is expected to provide a validation of the safety of the structural designs of the existing equipment. The results should indicate if the current designs are overly conservative and include excessive material usage or redundant structural features. Furthermore, the comparison of analytical solutions with the numerical simulation results is expected to verify the reliability of the FEA models.

The importance of this work lies in the potential of the data to support future design decisions for similar equipment, enabling more efficient and performance driven structures. Optimized

material usage would lower manufacturing costs for the company and, more broadly, contribute to improved environmental sustainability.

2 Methodology

This section outlines the procedure used to perform the structural assessment of each type of equipment. In general, the process begins with background research on the design principles of the equipment, followed by a description of the 3D modelling process, and concludes with the steps taken to conduct the static simulations using FEA software.

The fatigue evaluation follows a different approach, as it does not involve 3D modelling or FEA. Instead, it is based solely on analytical methods. Accordingly, this part is structured differently: it first presents the parameters of the fatigue study and then describes the analytical approach used for the fatigue assessment.

2.1 Pressurized vessel

The design of a pressurized vessel primarily focuses on its ability to withstand the internal or external pressure forces it encounters. There may be other considerations used in the design as well, which may influence its geometry or material selection (Spence and Tooth, 2014, 16). However, this thesis concentrates on the vessel's structural integrity under internal pressure. Since the equipment will operate under conditions involving a constant internal pressure, it is appropriate to use pressure vessel standards as a guideline for the design of the vessel's main body.

The design of pressure vessels follows strict standards set out in international codes, EN ISO 13445 in Europe and ASME Section VIII in the United States. These standards incorporate safety margins that account for the major primary loadings, as well as secondary localized stresses, for example those arising from the structural components of the pressure vessel. While the codes provide formulae for determining the required thickness of the main components, the design engineer must still apply their own judgement to ensure that the overall design is appropriate for its intended function (Moss and Basic, 2013, 2).

The required thickness of cylindrical shells is set out in EN ISO 13445 Clause 7.4.2 Cylindrical Shells (SFS, 2021)**.** The following formula is used to determine wall thickness:

$$e = \frac{P \cdot D_e}{2 \cdot f \cdot z - P} \quad (1)$$

Where:

e – Wall thickness (mm)

P – Design Pressure (MPa)

D_e – Outside diameter of shell (mm)

f – Nominal design stress (MPa)

z – Weld quality factor

When the thickness of the vessel wall is small relative to the radius (i.e. radius-to-thickness ratio ≥ 10), the vessel can be classified as thin-walled, which greatly simplifies the design calculations. Under this condition, the stresses caused by internal pressure are referred to as membrane stresses and are considered to be uniformly distributed through the thickness of the shell wall (Moss and Basic, 2013, 3).

A vessel subjected to internal pressure experiences a triaxial state of stress, namely:

σ_1 – Circumferential/ Hoop Stress

σ_2 – Longitudinal Stress

σ_3 – Radial Stress

For thin-walled vessels, the radial stress is negligible in comparison to the other two stresses, so it is not considered in the design calculations. So, the state of stress can be approximated as biaxial. The calculation of stresses alone is not sufficient to carry out the analysis. Once the stresses are determined, a failure theory must be applied that can point out when failure will occur or what thickness is required for the shell wall (Moss and Basic, 2013, 3).

In this thesis, the Maximum Principal Stress theory is adopted due to its simplicity and widespread use in industry (Moss and Basic, 2013, 4). For this theory, the stress in circumferential direction is the governing stress, thus it is used to determine the thickness of the shell:

$$e = \frac{P \cdot R}{\sigma_{max}} \quad (2)$$

Where:

R – Radius of shell (mm)

σ_{max} – Maximum allowable stress (MPa)

The resulting thickness obtained from the standards will be compared to the result from the Maximum Principal Stress theory. These results will then be compared to the wall thickness specified in the engineering drawings provided by the company to assess if the vessel is overdesigned. Subsequently, a static simulation will be conducted to determine the extent to which material usage can be reduced while still maintaining an acceptable margin of safety.

2.1.1 3D Design

Prior to the performing the static simulation, a 3D model of the pressurized vessel was developed. Using reference drawings provided by the company, each part was designed and assembled in a 3D CAD software. The modelling approach for simulation purposes differs from that used for manufacturing design; accordingly, the geometry was simplified by removing non-essential features such as bolt holes. Contacting surfaces between components were carefully defined to ensure proper alignment, with the Cavity feature used extensively for this purpose. This approach ensured that no unintended interferences or gaps were present in the model, which could otherwise impact the accuracy of the simulation results.

2.1.2 Finite Element Analysis

After finalizing the 3D model, the file was saved in Parasolid format (.x_t) for easy readability in the simulation software.

In FEA, a structure can be analyzed using various element types, such as solid, plate, rod, and shell elements. For thin structural components, plate and shell elements can be utilized. Plates are used for thin members which are flat and experience only in-plane loads. Shell elements are suitable to use when the geometry is curved and is subjected to both in-plane

and out-of-plane loads. Therefore, to solve the stresses in three dimensions, shell elements are generally applied (Madenci and Guven, 2015, 355).

These types of elements are preferred over solid elements for thin structures because they significantly reduce computation time while maintaining a comparable level of accuracy. Modelling a thin structure with solid elements requires a very fine mesh, including at least two elements through the thickness, which substantially increases the computational cost. In contrast, with shell elements, the thickness is not explicitly shown in the model but is specified by the user as a property of the element. This not only reduces computational effort, but it is also much easier to test different wall thicknesses during the analysis.

The pressurized vessel consists mainly of several thin, curved components, so shells were the ideal element type for this analysis. Other thicker components such as the leg supports, remained as solid elements. The interactions between these elements also had to specified accordingly. The detailed simulation procedure is described below:

1. Once a Static Structural module was set up, the cylindrical shell, ellipsoidal head and the accept and outlet pipes were converted into surface bodies using the Midsurface function. In order to create a conformal mesh, the surface bodies were connected using the Shared Topology function. This ensures that the bodies are connected at the nodal level, resulting in a continuous mesh across the interfaces.

2. Next, the material was assigned for all bodies and the mesh was generated. The material used was EN 1.4301, a general-purpose stainless steel known for its good corrosion resistance and weldability characteristics. For this thesis, conservative values for its Yield Strength (190 MPa) and Ultimate Strength (500 MPa) were used (Sverdrup Steel, ,n.d.) to conform with safe engineering practices.

3. The assembly was now discretised using a mesh containing a combination of solid elements and shell elements, where shell elements are applied to the surface bodies defined in step 1. For shell elements, the z-axis is always normal to the element surface, however the x and y axes may not be aligned correctly. To accurately extract the hoop and longitudinal stresses, the x and y axes can be specified using the Element Orientation function.

4. Once the mesh was defined, the contacts between different components of the assembly were specified as Bonded, since all parts are welded together. Contact definitions were

not required for the various surface bodies since they already shared a conformal mesh. Apart from defining contacts within the leg support assembly, the interface between the shell and solid components required special attention. To ensure compatibility of nodal deformations between shell and solid elements, the contact formulation was adjusted accordingly. A Multi Point Constraint (MPC) was used for the Contact Formulation, and "Projected, Displacement Only" was selected for the Contact Constraint Type. These modifications connect the displacements of the contacting bodies together.

5. The final step was to set up the boundary conditions of the model. A fixed support was set at the base of the feet. The loading conditions included pressure on the vessel walls and standard earth gravity. A minor modification was required for the pressure loading. Since the design pressure is applied at the inner surface of the vessel, it was scaled to account for application on the midsurface of the vessel. This adjustment is necessary because the midsurface has a larger area than the actual inner surface of the vessel; therefore, the applied pressure value was corrected as follows:

$$P_1 A_1 = P_2 A_2$$

$$P_2 = P_1 \times \frac{A_1}{A_2} \tag{3}$$

6. For this vessel, a physical test was conducted by the company at their premises on February 6th, 2026 to ensure that it could withstand the design pressure with an additional safety margin. Micrometer gauges were placed at three specified locations on the vessel to measure the deformation as the internal pressure was increased in small increments. The deformation results obtained from the static simulation were compared to the result from the pressure test to validate the numerical model.

2.2 Large Non-Pressurized Mixing Vat

Process equipment under atmospheric (non-pressurized) loading can also be dimensioned using membrane stresses to determine sufficient wall thicknesses. The membrane stress state only exists in the ideal cylindrical or conical parts of the equipment, so any local bending stresses that might occur due to any geometrical irregularities are excluded from the membrane stress theory. Consequently, the wall thicknesses calculated are only applicable

to the main body of the vat, while other parts such as nozzles and supports need additional analyses (Nagy, 2023, 35).

The equipment being analysed in this section is a large tank (volume greater than 100m^3) used to mix materials in liquid form using a rotor located at the bottom of the vessel. As this is a static analysis, the main loading component is hydrostatic pressure caused by the liquid. So, the analytical calculation assumes that membrane stress caused by the hydrostatic pressure is the governing factor for the determination of the vat wall thickness. By doing so, catastrophic failure of the vessel is avoided as it is ensured that the vessel will not burst apart under the hydrostatic load (Nagy, 2023, 36).

A hydrostatic load requires a relatively more complex model since it varies along the height of the vessel. The tank being analyzed consists of three bodies welded together – the main cylindrical wall, a conical bottom inclined at roughly 50° from the vertical wall and a second conical section below it inclined at approximately 25°. Existing literature provides a formula for a simplified structure which consists of a single conical section connected to the vertical cylindrical wall. For a tank where the height of the wall is greater than the height of the cone, the most likely point of failure is the junction of the cone and cylinder (Nagy, 2023, 53).

The formula below calculates the minimum thickness of the conical section based on a hydrostatic pressure exerted from the level of liquid above it.

$$s \geq \frac{p_{hid} \cdot R}{\sigma_{allowed} \cdot \cos \alpha} \tag{4}$$

Where:

s – minimum wall thickness (m)

p_{hid} – ρ (Water density) (kg/m^3) x Gravity (m/s^2) x H1 (Height of liquid) (m)

R – Radius measured up to the middle of the wall thickness

$\sigma_{allowed}$ – Allowable stress (Pa)

α – Angle measured from vertical cylinder wall to the cone wall

The dimensions are represented in Figure 2 below.

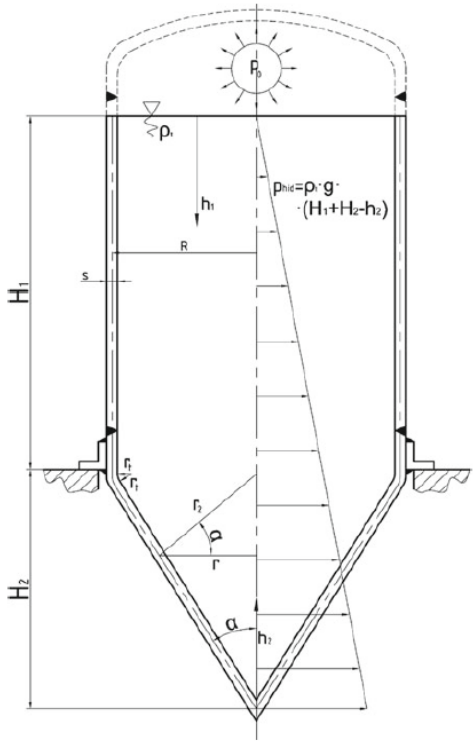


Figure 2 - Cylindrical tank with conical end (Nagy, 2023, 51)

The cone wall thickness calculated from the analytical calculation was compared to the value in the company reference drawing to assess how much the wall is over dimensioned. Thereafter, a static simulation was conducted to determine a wall thickness value that is within a reasonable factor of safety.

2.2.1 3D Design

Following the same procedure as the pressurized vessel, a 3D model was created using company reference drawings prior to performing the static simulation. Once again, the Cavity feature was helpful in ensuring the model had no gaps or interferences that would affect the simulation.

2.2.2 Finite Element Analysis

The simulation steps were mostly identical to the pressurized vessel. The cylindrical wall, support rings, conical sections and baffles were modelled using shell elements as these

components can be characterized as thin structures. Once again, the leg supports were the main solid bodies in the model. Contacts between the various components were defined, material was assigned and boundary and loading conditions were applied.

An extra step was required in this analysis due to restrictions on the size of the permitted numerical model in the student version of the simulation software. As the structure is quite large, the total number of finite elements of the various components exceeded the allowable limit. Firstly, the structure was analyzed without the use of a top cover to measure the stress and deformation without the additional support provided by the cover. Then, the analysis was repeated with the top cover included in the assembly. Since this increased the number of elements beyond the allowed limit, the use of the Symmetry tool was necessary to reduce the size of the model.

If the geometry of a structure exhibits symmetry, then it can be computationally less expensive to carry out the analysis on only one section of the structure that represents the entire physical system. This tool can only be used if the following features of the model are symmetrical (Madenci and Guven, 2015, 20):

- Geometry
- Material
- Loading conditions
- Constraints applied

Since the equipment being analyzed is a cylindrical vessel under hydrostatic pressure, it is reasonable to assume symmetrical conditions. The steps are detailed below:

1. Using a static structural model, the model geometry first had to be modified to create the symmetrical region. To do this, symmetry planes were created to split the geometry using a cutting tool. Since the tank is axisymmetric, planes were created at angles of 120°, thus splitting the structure into three equal regions. As only one region is required for the analysis, all the components in the other two regions were excluded for the purposes of the simulation.
2. Once the geometry was split, the resulting model was used for simulation. Since only a portion of the model was under analysis, the number of contacts was significantly

reduced. Symmetry Regions were then applied at the symmetry faces created for each component in the model. The Symmetry Regions constrained the translational and rotational degrees of freedom along the specified faces. Local co-ordinate systems were created at each face, with the Z-axis aligned normal to the faces to ensure that constraints were properly oriented.

3. At this stage, the Symmetry tool was applied by selecting the symmetrical faces and defining the Symmetry Normal along the Z-axis, which corresponds to the normal direction specified in the Symmetry Regions. This approach ensured the structure would respond appropriately as an accurate representation of the full model.
4. The loading for the symmetrical model needed to be appropriately adjusted. The hydrostatic pressure and gravity loads were applied unmodified, while any forces acting on the structure were reduced by a factor of 3 since only a third of the full model was under analysis. Once loading was correctly applied, the simulation was ready to run.

The symmetrical model was important to include in the study as the cover is considered a standard component of larger vats. The inclusion of the cover in the study leads to more realistic results.

2.3 Non-pressurized Mixing Vessel

This equipment consists of a rotor mounted in a horizontal orientation within a small vessel and is used to break down material to finer levels. The fatigue study carried out for this equipment focused solely on the bolts and pins that connected the spline to the rotor. The various components and overall assembly are illustrated in Figure 3 below. In the actual equipment, a shaft would be connected to the spline that would drive the rotor at high rotational speeds.

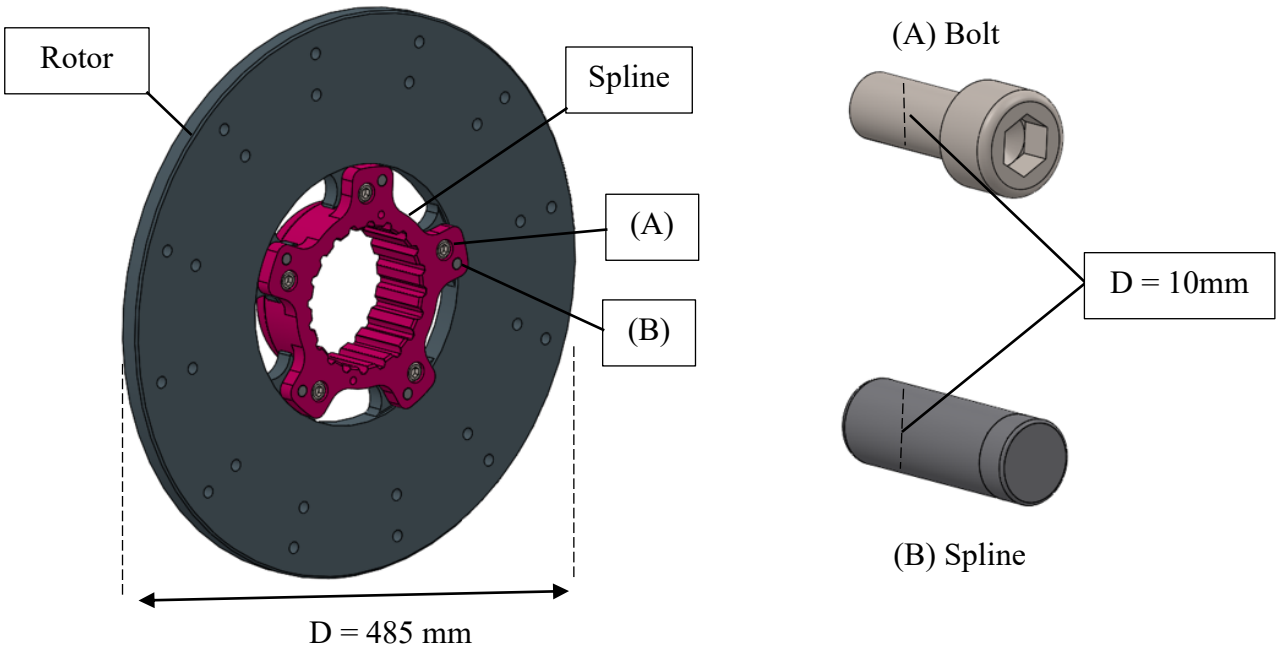


Figure 3 - Rotor Assembly

The assembly consists of five bolts and pins that connect the spline to the rotor. Each bolt undergoes single shear stress across the shank cross-section, preventing the spline from separating from the rotor during the equipment operation. The goal is to determine the torque amplitude that would lead to a fatigue failure of the bolt connection, given a specified finite life and assuming the connection is experiencing cyclic loading. To carry out this task, a traditional nominal stress approach was used.

Fatigue failure is the one of the most common causes of structural failure during normal operations. It is particularly dangerous because it is difficult to detect, often occurring at stress levels that are lower than the elastic strength of the material, but the failure occurs quickly. This type of failure arises when a structure is experiencing repeated cyclic loading and manifests as small cracks developing in the structure, which grow as the loading continues. In actual conditions, the loading is normally never perfectly cyclic and can be often variable in nature (Bathias and Pineau, 2010, 8). However, for this study, a simplified model is used where it is assumed that the loading is perfectly cyclical in order to arrive at an estimated value for the stress range experienced by a single bolt.

EN 1993-1-9 provides guidelines for assessment of fatigue resistance of different types of members that experience cyclic loading. Section 7.1 provides a general formula for calculating fatigue strength for a constant amplitude loading:

$$\Delta\sigma_R^m \cdot N_R = \Delta\sigma_C^m \cdot 2 \times 10^6 \quad (5)$$

Where:

$\Delta\sigma_R$ – stress range corresponding to N_R cycles (MPa)

N_R – design life in number of cycles using a constant stress range

$\Delta\sigma_C$ – reference fatigue strength at 2×10^6 cycles (MPa)

m – gradient of S-N curve

In this case, we are trying to figure out the stress range $\Delta\sigma_R$ given N_R. Thus, the formula can be re-arranged as:

$$\Delta\sigma_R = \Delta\sigma_C \times \left(\frac{2 \times 10^6}{N_R}\right)^{1/m}$$

As per section 7.1 (4), the reference fatigue strength $\Delta\sigma_C$ can be obtained from Tables 8.1 – 8.10 in the standard, which provides the fatigue (FAT) classes for different members including bolts, thus the formula is rewritten as:

$$\Delta\sigma_R = FAT \times \left(\frac{2 \times 10^6}{N_R}\right)^{1/m}$$

If the component is thicker than 25mm, then a stress concentration factor k_s is applied to the FAT class. Section 3 of EN 1993-1-9 provides two options for fatigue assessment method: the damage tolerant or the safe life method. The damage tolerant approach provides an acceptable reliability for components that are designed to undergo routine maintenance to check for fatigue damage throughout the design life. In this case, if fatigue damage were to occur, then the load would be distributed to other members in the assembly. On the contrary, the safe life method assumes that in-service inspections for detecting fatigue damage would not be performed, and crack formation in one component would quickly lead to failure of the entire assembly. Based on the chosen assessment method, a partial safety factor γ_{Mf} is applied to the stress range $\Delta\sigma_R$. Using these modifications, the formula can be represented as:

$$\Delta\sigma_R = \frac{k_s \times FAT}{\gamma_{Mf}} \times \left(\frac{2 \times 10^6}{N_R}\right)^{1/m} \quad (6)$$

Using the obtained stress range $\Delta\sigma_R$, the applied forces on each bolt can be deduced using:

$$\Delta\sigma_R = \frac{F_{max} - F_{min}}{A} \tag{7}$$

Where A is the cross-sectional area of a single bolt. Once the minimum and maximum forces on a single bolt are obtained, the torque range can also be calculated:

$$\tau = r \times F \tag{8}$$

Where:

τ – Torque (Nm)

r – distance from the centre of the shaft to bolt (m)

3 Results

This section presents the results of the analytical calculations, 3D modelling and static structural simulations for the pressurised vessel and mixing vat. Finally, the results of the fatigue analysis for the non-pressurized mixing vessel are provided.

3.1 Pressurized vessel

This section focuses on the design and analysis of the pressurised vessel. The assessment begins with analytical calculations based on relevant design standards to determine the required wall thickness.

3.1.1 Analytical solution

Using the formula provided by the standard EN 13445, the thickness of the shell is calculated as follows.

- As per company sources, the maximum design pressure for the vessel is 0.35 MPa.
- f (Nominal design stress) is calculated as:

$$f = \min\left(\frac{R_{p0.2}}{1.5}, \frac{R_m}{2.4}\right) \tag{4}$$

$$f = \min\left(\frac{190\ \text{MPa}}{1 \cdot 5}, \frac{500\ \text{MPa}}{2 \cdot 4}\right) = 126.7\ \text{MPa}$$

- z (Weld quality factor) is assumed to be 1 (indicating a high-quality weld).
- D_e is 1700mm as per reference drawings.

Plugging in the values into the thickness equation (1) provided by the standard:

$$e = \frac{0.35\ \text{MPa} \cdot\ 1700\ \text{mm}}{2 \cdot 126 \cdot 7\ \text{MPa}\ \cdot 1 - 0 \cdot 35\ \text{MPa}} = 2.35\ \text{mm}$$

Using the Maximum Principal Stress theory, equation (2), the thickness is calculated as:

$$e = \frac{0.35\ \text{MPa} \cdot 850\ \text{mm}}{126.7\ \text{MPa}} = 2.35\ \text{mm}$$

3.1.2 3D Model

This section illustrates the 3D model that is used for the static simulation. Components have been designed using reference drawings provided by the company including slight adjustments to simplify the model for simulation purposes.

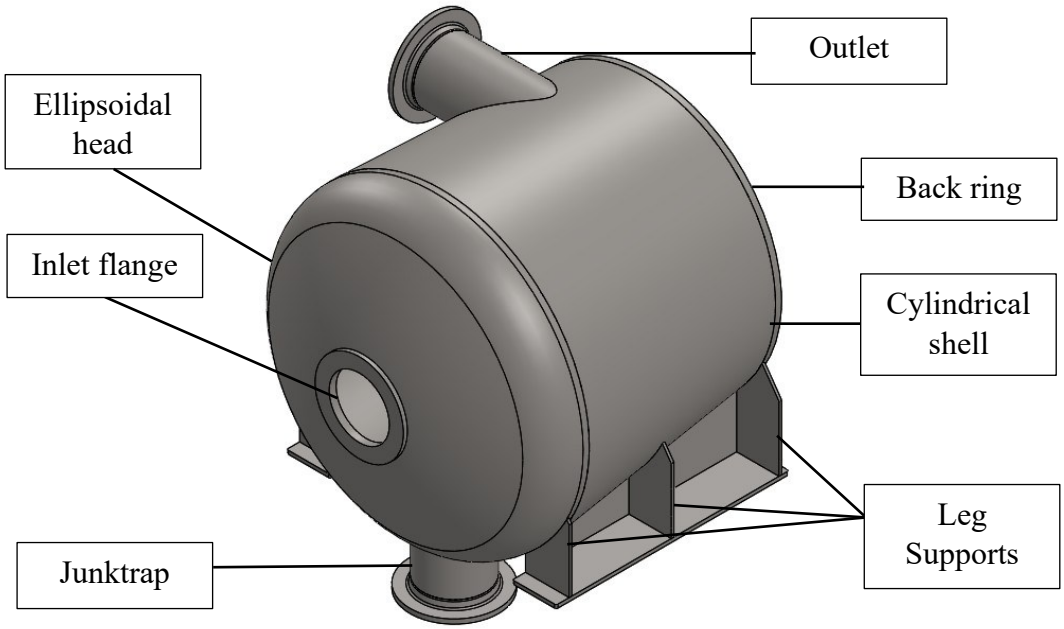


Figure 4 – Fully Assembly Model

Figure 4 shows the main components of the equipment. As per original drawings, the thicknesses of the components in the model are given in the table below.

Table 1 - Components Plate Thickness

Component	Thickness (mm)
Ellipsoidal head	12
Inlet flange	40
Junktrap	4
Outlet	4
Back ring	35
Cylindrical shell	12
Leg Supports	15

These plate thicknesses will be used for the initial static assessment.

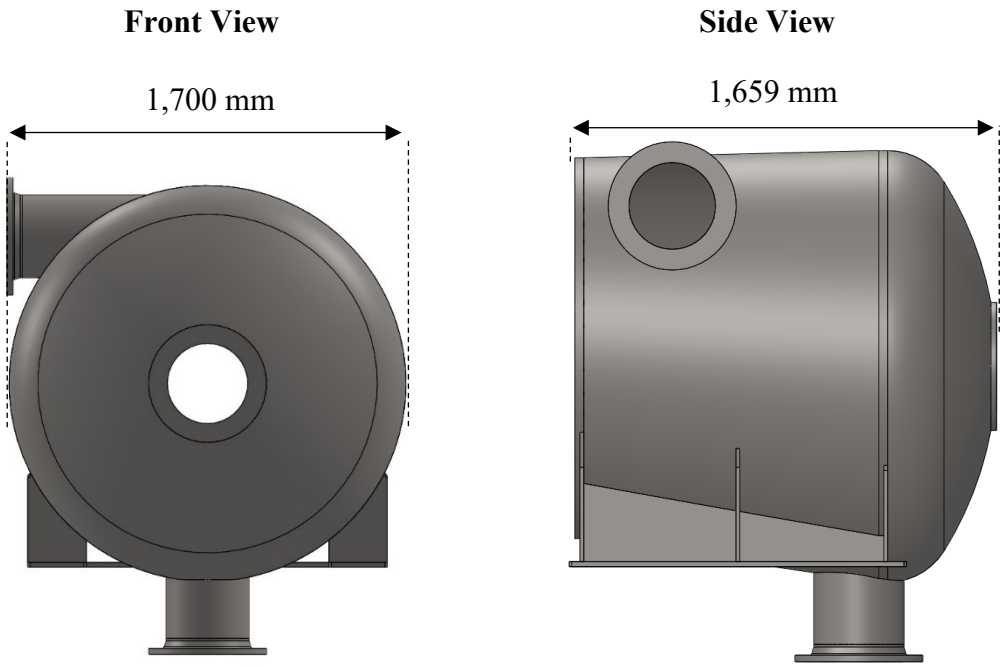


Figure 5 – 3D Model views

Using the dimensions shown in Figure 5 and plate thicknesses in Table 1, the total mass of the equipment is approximately 1.48 tonnes.

3.1.3 Stress Analysis

As mentioned in section 2.1.2, the first step to conduct the static simulation was to create the midsurfaces. The transparent bodies shown in Figure 6 represent the shell surfaces that were created using the Midsurfacing tool.

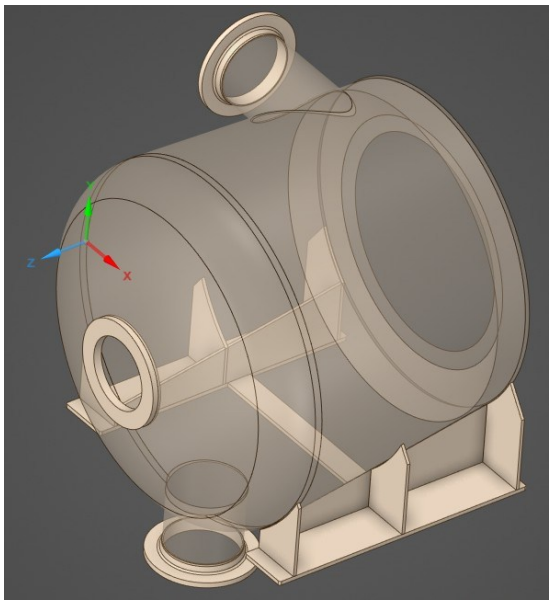


Figure 6 – Midsurfacing

Once the midsurfaces were defined, the mesh was created. The mesh displayed in Figure 7 consists of a combination of shell elements (cylindrical shell, ellipsoidal head and the inlet and outlet pipes) and solid elements (leg supports and flanges).

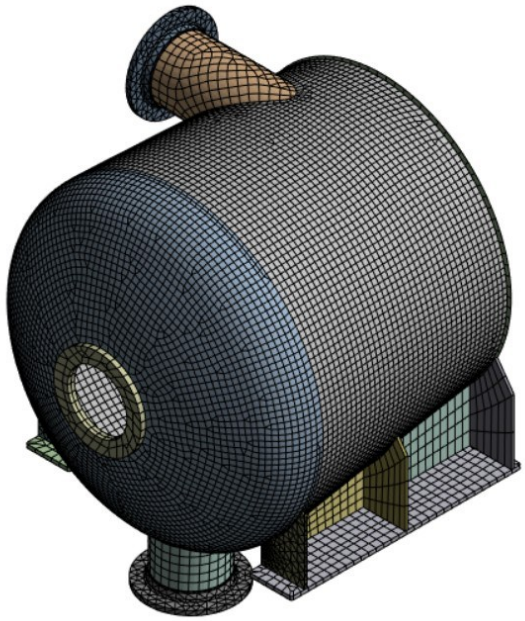


Figure 7 – Meshing

Figure 8 shows a section of the shell elements of the cylindrical shell of the vessel. The blue lines represent the z-axes of each element, while the red and green lines represent the x and y axes respectively. As shown, the red lines follow the circumferential direction of the shell while the green lines are aligned along the length. We can use this orientation in the post-processing of results to extract stress in the direction of each axis (x axis for circumferential stress and y axis for longitudinal stress).

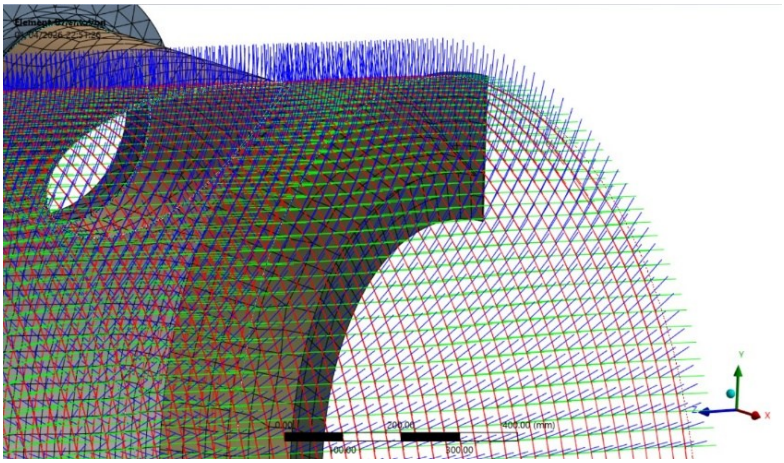


Figure 8 - Element Orientation

Once the surface model was ready, the boundary conditions could be specified. Figure 9 shows the fixed supports, gravity and pressure loadings.

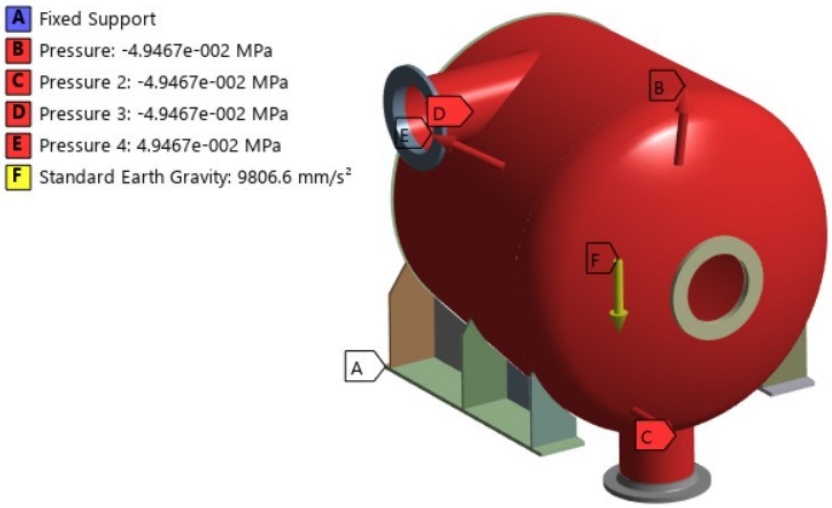


Figure 9 - Boundary Conditions

The pressure loading used in the static simulation was defined to reflect the physical test conditions. According to the company, the maximum operating pressure of the equipment is 0.35 MPa. For testing purposes, a safety factor of 1.43 was applied, resulting in a test pressure of 0.5 MPa. The pressure was increased in increments of 0.05 MPa, and deformation was recorded at each interval at the three micrometer gauge locations.

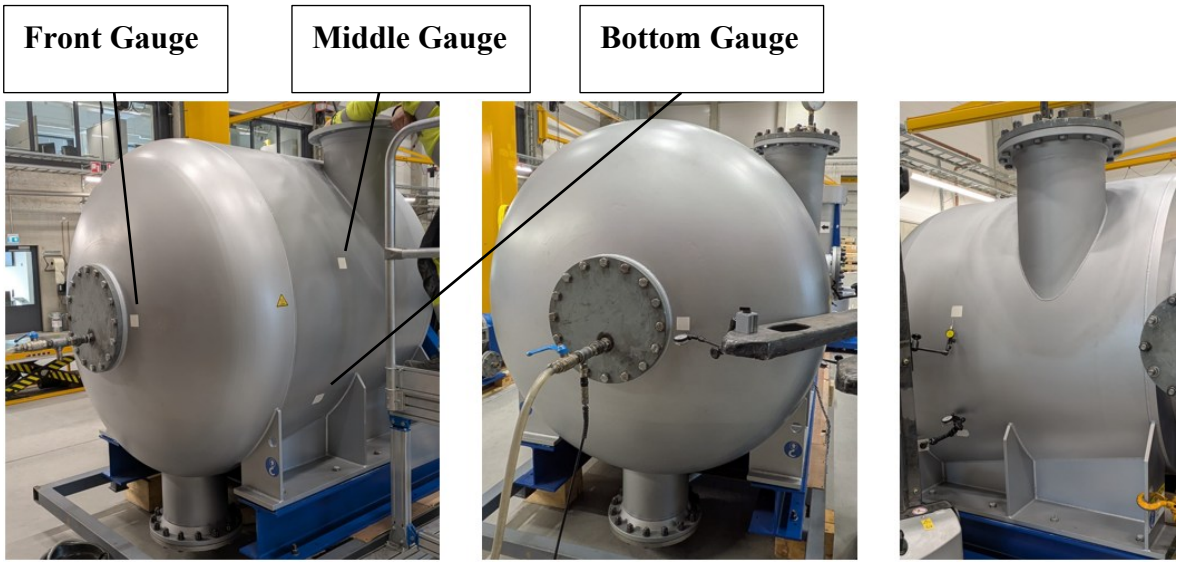


Figure 10 - Micrometer gauge locations

Figure 10 shows the three locations where the micrometer gauges were installed during the physical test. Each gauge measures the deformation of the wall in the direction normal to the surface. A resolution of 0.01 mm allows accurate measurement of very small deformations.

The same loading conditions used in the physical test were applied in the static simulation. Firstly, the pressure was adjusted using equation (3).

$$P = 0.5005\ \text{MPa} \times \frac{8.4998\ \text{m}^2}{8.6\ \text{m}^2} = 0.49456\ \text{MPa}$$

This pressure was applied in 10 load steps, corresponding to increments of 0.049 MPa, reflecting the physical test procedure. Using the Directional Deformation option in the Solution panel, the deformation was captured at specific nodal locations in the model, corresponding to the micrometer gauge positions used in the experimental test.

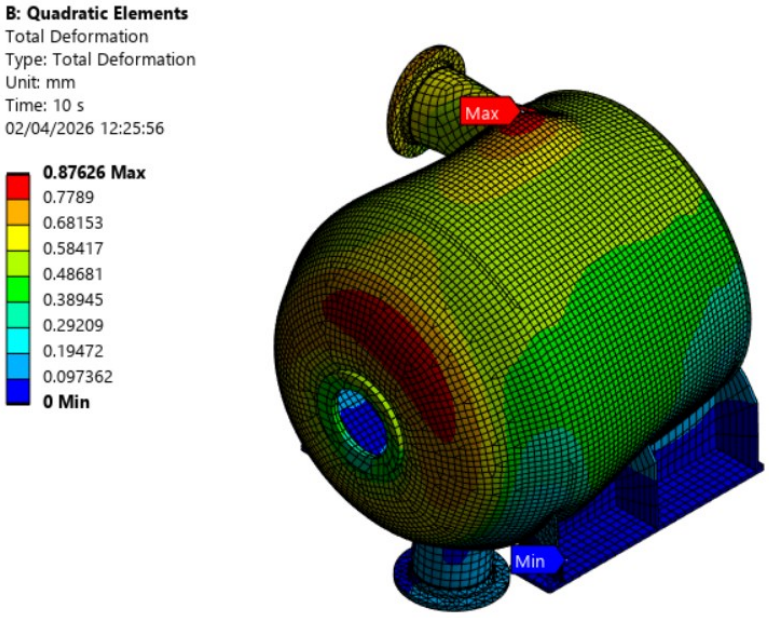


Figure 11 - Total Deformation

Figure 11 shows the total deformation of the model. The maximum occurs the top of the vessel, while negligible deformation is observed at the support legs.

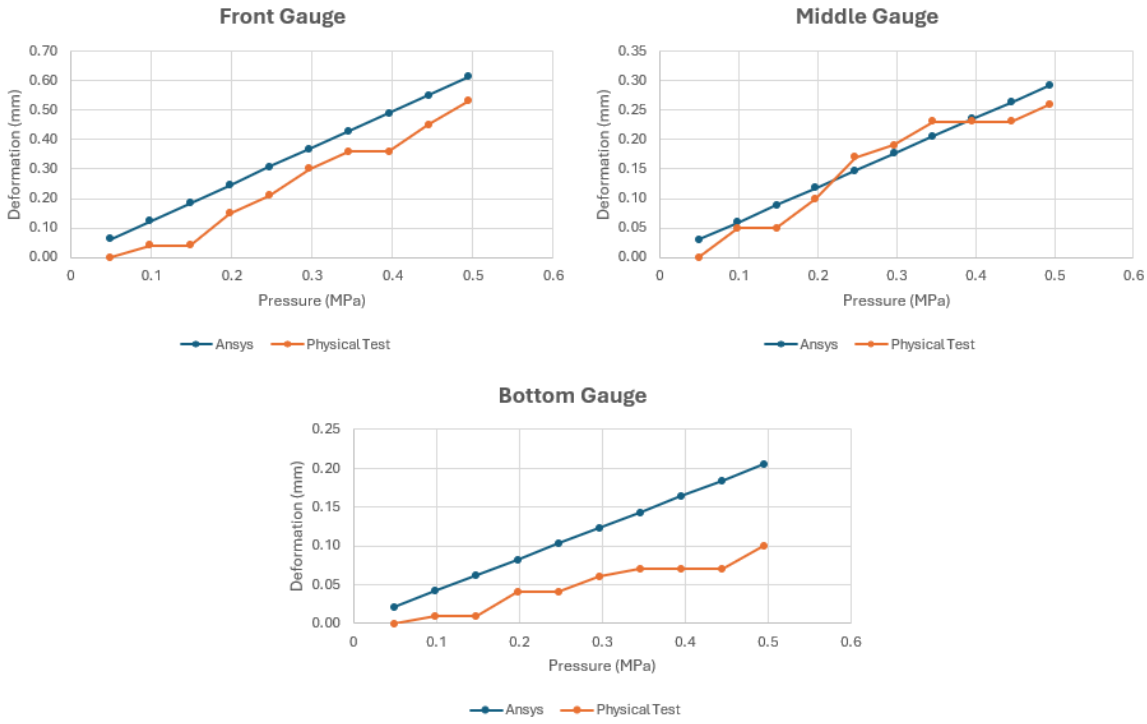


Figure 12 - Deformation Results

Figure 12 shows a comparison of the deformation results between the physical test and the static simulation. Since the simulation is a linear static analysis, the deformation increases linearly with applied pressure.

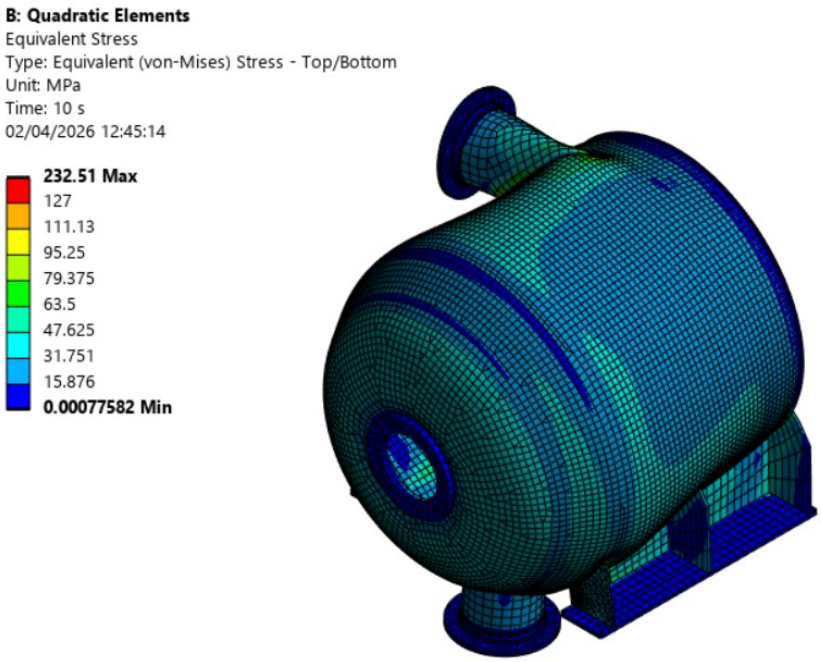


Figure 13 – Von Mises Stress Distribution

Using a material yield strength of 190 MPa and a safety factor of 1.5, the stress distribution legend was adjusted to highlight regions exceeding 127 MPa (190/1.5) in red. As shown in Figure 13, the majority of the vessel is subjected to relatively low stress levels, with a maximum stress of approximately 48 MPa indicated in the blue regions.

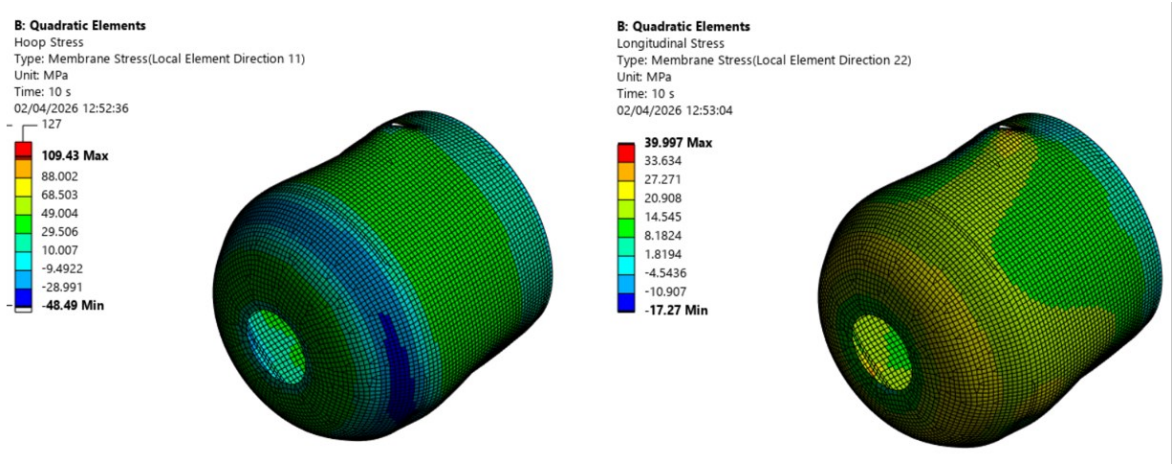


Figure 14 - Hoop Stress & Longitudinal Stress

Figure 14 shows the circumferential and longitudinal stresses resulting from the applied loading. Care must be taken when interpreting the results, as the legend scales differ between the two plots, meaning that identical colours correspond to different stress magnitudes. As expected from pressure vessel theory, the circumferential stress is the dominant stress component, with values ranging between 30 and 50 MPa over the majority of the vessel. The longitudinal stress is significantly lower in comparison. No regions exceed the allowable stress limit of 127 MPa.

Subsequent simulations were carried out using 6mm and 4mm vessel wall thickness.

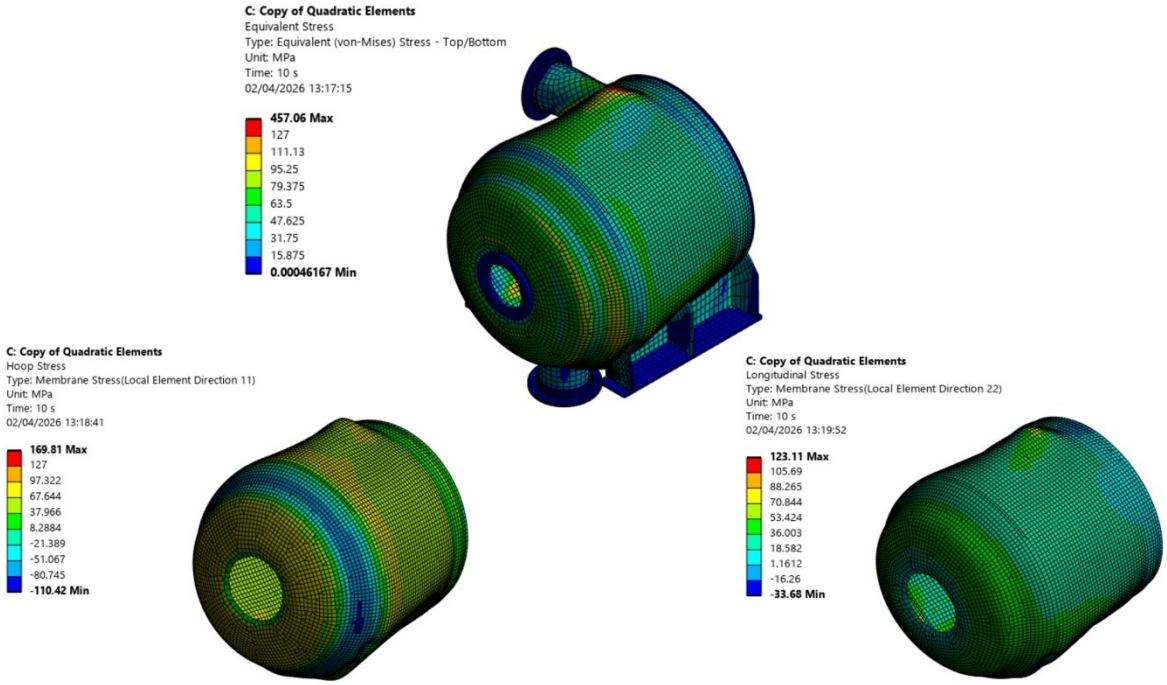


Figure 15 - Stress Distribution (6mm wall thickness)

Figure 15 depicts the stress distribution at maximum pressure for a wall thickness of 6 mm. The regions in red (which show a breach of the factor of safety limit) are all concentrated at the joint where the outlet pipe is welded to the main cylindrical vessel.

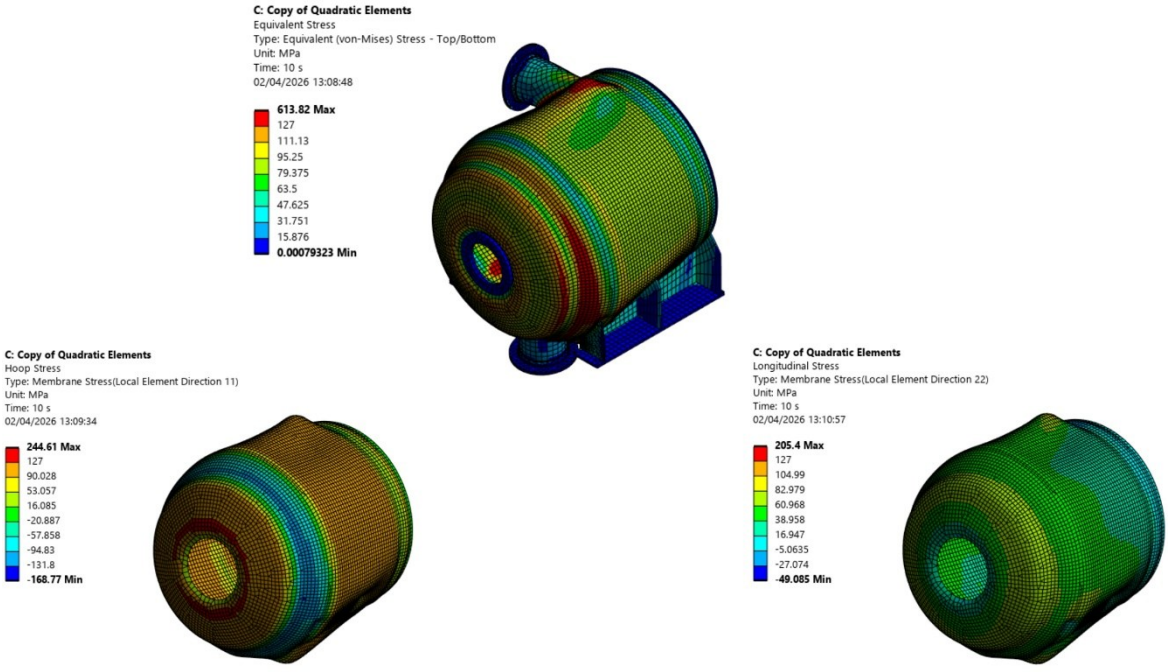


Figure 16 - Stress Distribution (4mm wall thickness)

Figure 16 shows the stress distribution in the vessel using 4mm wall thicknesses, corresponding to the same thickness as the pipe connections. Regions close to or exceeding the allowable stress (indicated in orange and red) are now more widespread and extend into the main body of the vessel, rather than being concentrated solely at the pipe connection as observed previously.

3.2 Large Non-Pressurized Mixing Vat

This section focuses on the design and evaluation of the non-pressurised mixing vat. The assessment begins with analytical calculations aimed at determining the required wall thickness. Following the analytical calculation, the results of the static simulation are presented.

3.2.1 Analytical solution

Using equation (4), the minimum wall thickness is calculated using the following parameters:

ρ – 1,000 kg/m^3

H1– 4.145 m

R – 2.9475 m

$\sigma_{allowed}$ – 127,000,000 (Pa) (Since the same material, EN 1.4301, is used for this structure as for the pressurized vessel, the allowable stress is kept the same to remain consistent in the methodology).

α – 40°

Therefore, minimum wall thickness for the conical section is calculated as:

$$s \geq \frac{\left(1000\,\frac{\mathrm{kg}}{\mathrm{m}^3} \times 9.81\,\frac{\mathrm{m}}{\mathrm{s}^2} \times 4.145\right) \times 2.9475\ \mathrm{m}}{127{,}000{,}000\,\frac{\mathrm{N}}{\mathrm{m}^2} \times \cos(40°)} = 0.00123\ \mathrm{m} = 1.23\ \mathrm{mm}$$

3.2.2 3D Model

The section outlines the various components of the full assembly of the mixing vat. Minor adjustments are made to the geometry to ensure a smooth simulation.

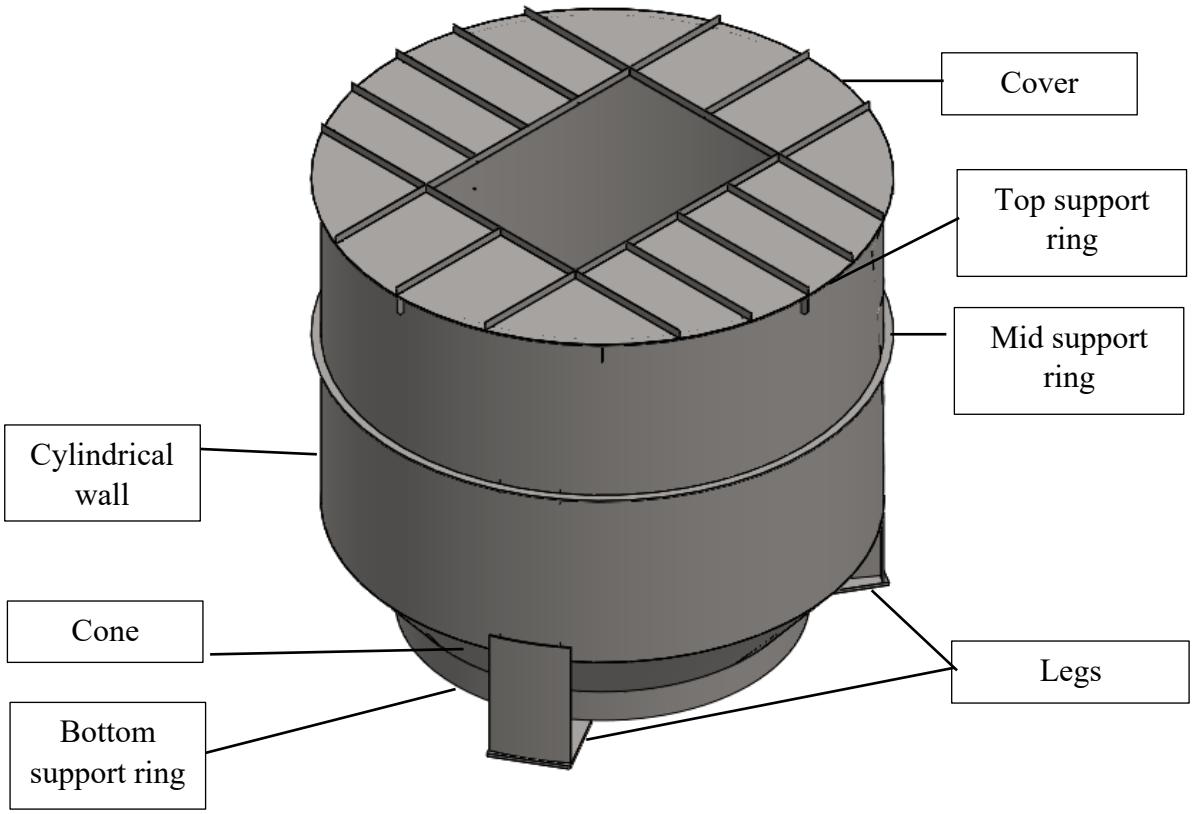


Figure 17 – Full Assembly Model

Figure 17 shows the major components of the mixing vat. The conical bottom is divided into two sections, both with a thickness of 8mm. The cylindrical wall is 5mm.

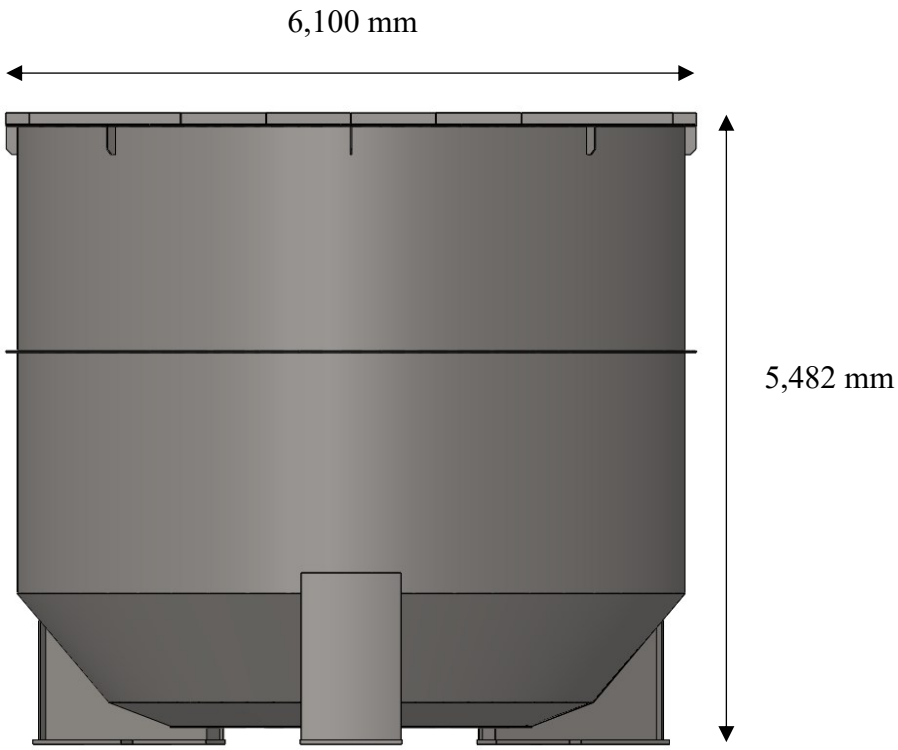


Figure 18 - Side View

The full assembly weighs approximately 9 tonnes without the top cover (≈ 9.8 tonnes with the cover included). Figure 18 shows the main dimensions of the structure. In this figure, the bottom ring has been hidden to display the two angled sections of the bottom conical cone.

3.2.3 Stress Analysis

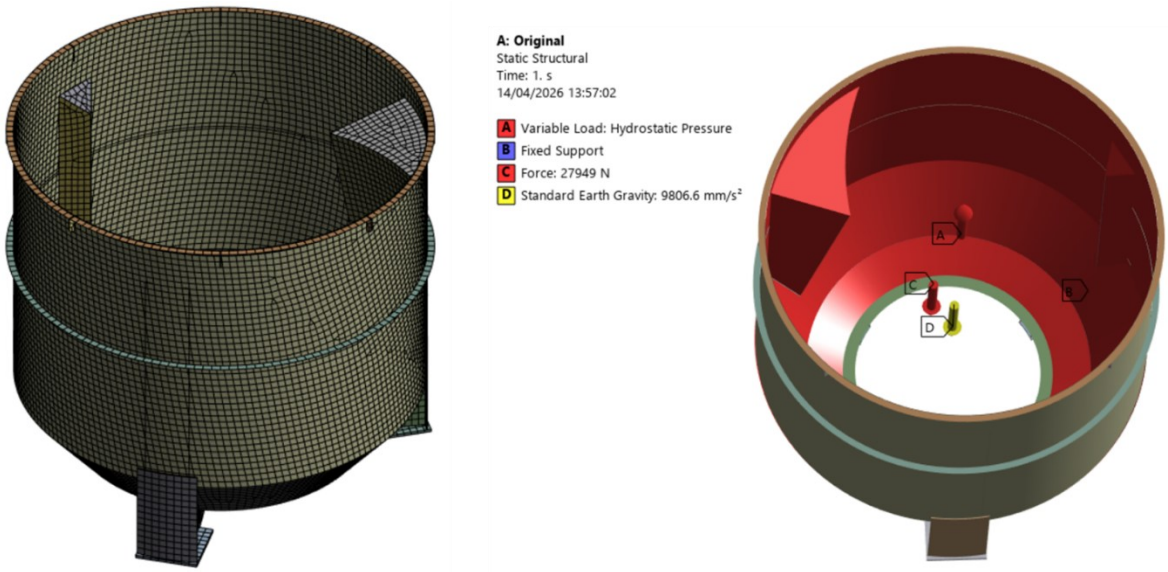


Figure 19 - Mesh, Boundary Conditions and Loading

Figure 19 shows the applied mesh and boundary conditions for the model. The hydrostatic pressure was applied to all inside surfaces, indicated in red. The mass of the rotor (approximately 2,850kg) was applied to the face of the bottom ring. Fixed supports were applied at the bottom of the leg supports. Finally, gravity force was included to represent real conditions.

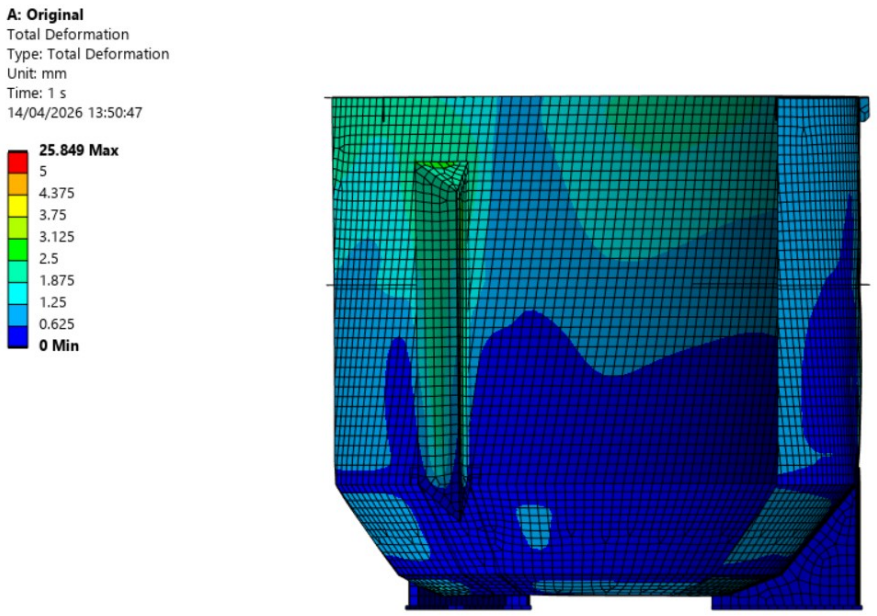


Figure 20 - Total Deformation (8mm cone wall thickness)

Using reference wall thickness values, the average deformation on the cylindrical wall was 1.4mm, while the conical section experienced approximately 0.5mm of deformation. As depicted in the section view in Figure 20, the deformation increased with distance from the leg supports, reaching an average of 1.9mm at the top ring support.

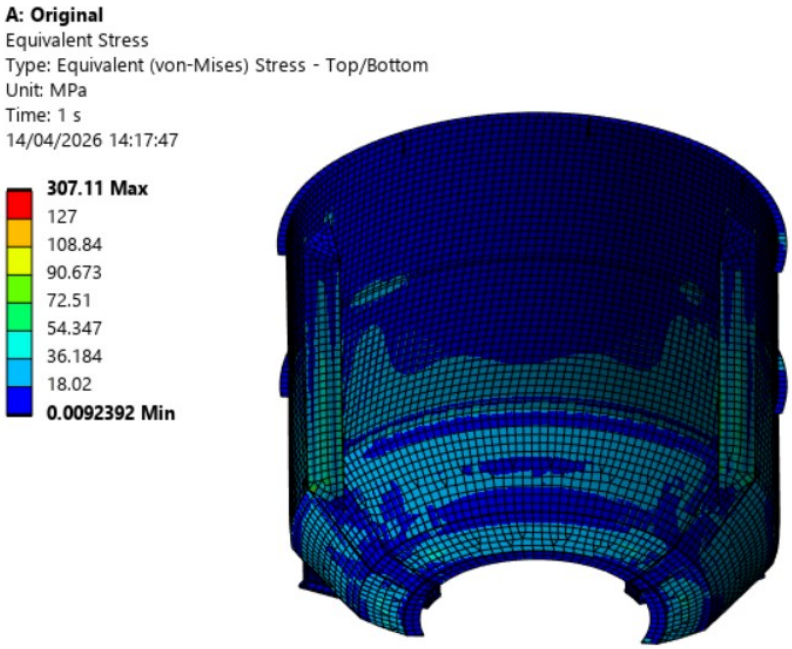


Figure 21 - Stress Distribution (8mm cone wall thickness)

As before, a stress limit of 127 MPa was used for the analysis. As shown in the figure above, the blue regions indicate that stress values are well within the safe region. Areas in light blue on the conical section experience stress of approximately 20 MPa and below. The junction of the wall and cone shows the highest stress values, approximately 50 MPa. However, stress levels decrease when approaching close to the baffles. The leg supports also experience low stress levels, averaging 14 MPa. The baffles experience slightly higher stress, averaging around 27 MPa.

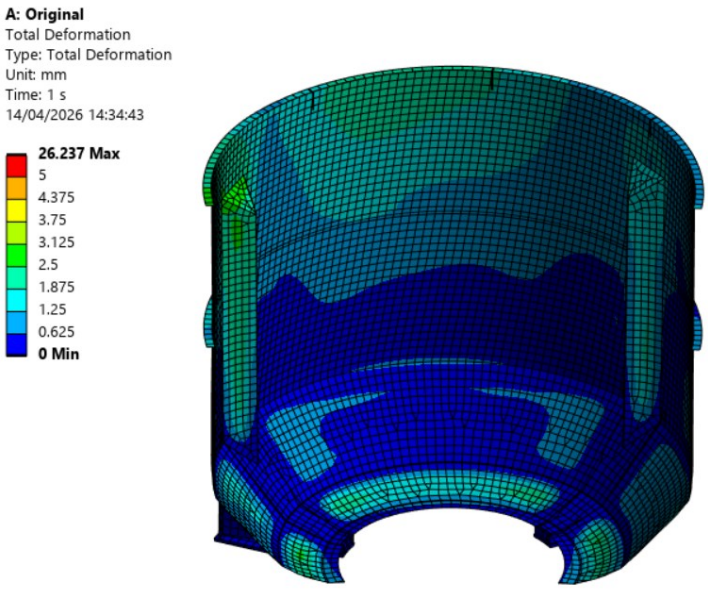


Figure 22 - Total Deformation (5mm cone wall thickness)

Using 5mm wall thickness for the conical bottom section, the deformation increases slightly in the bottom cone. The green highlighted areas at the bottom show deformation of approximately 1.7mm. Deformation for the cylinder and top support remain largely the same.

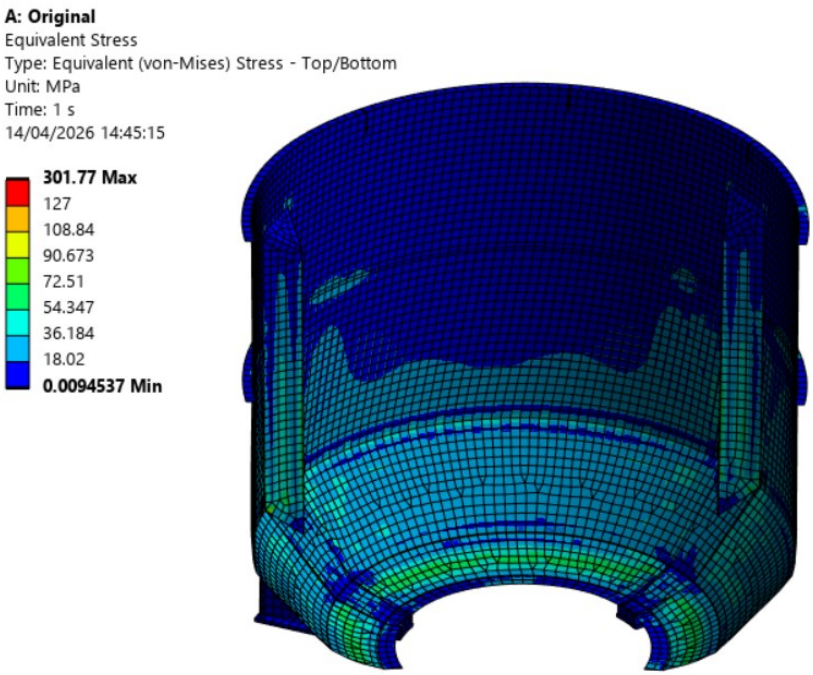


Figure 23 - Stress Distribution (5mm cone wall thickness)

Average stress values increase slightly but remain below 127 MPa. Average stress in the conical section increases to approximately 30 MPa, while the stress in the cylinder remains mainly unchanged at 16 MPa. Reducing the thickness of the conical section to 5mm results in a material saving of approximately 630kg, which represents a 7% material saving for the total structure.

Finally, the symmetry study was conducted to measure the effect of the cover on the stiffness of the structure.

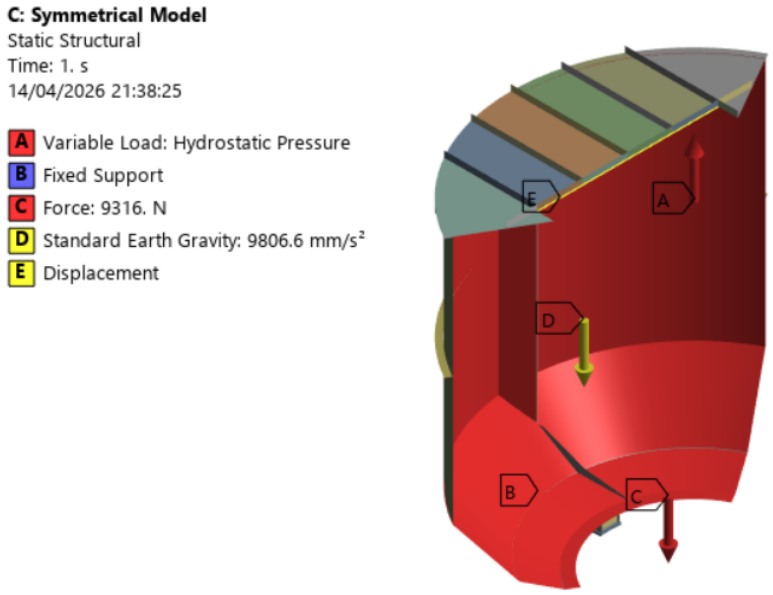


Figure 24 - Symmetrical Model

Figure 24 shows the loading conditions for the symmetrical model. The mass of the rotor has been divided by 3 to obtain the adjusted force value, while the other loads stayed unchanged. Wall thicknesses were as per original company drawings (8mm for bottom sections).

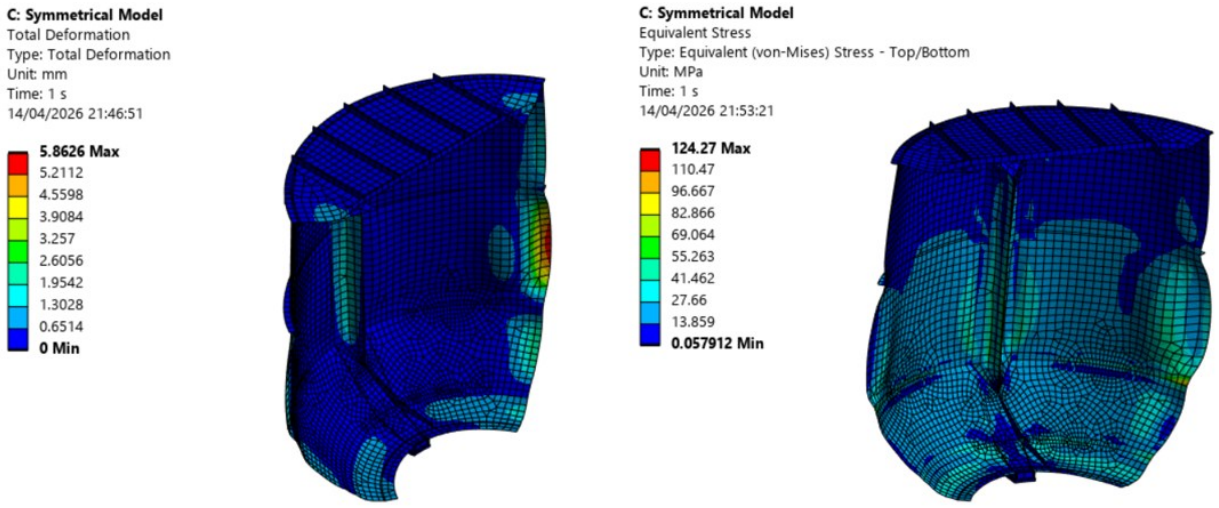


Figure 25 - Deformation and Stress Analysis (Symmetrical model)

With the inclusion of the cover, the deformation drops to negligible levels. Average deformation on the wall and bottom cone was 0.44mm, while deformation at the top ring was 0.16mm. The stress on the conical sections and cylinder remained largely the same (between 15 to 20 MPa), while stress on the baffles decreased slightly from 27 MPa to 24 MPa.

3.3 Non-pressurized Mixing Vessel

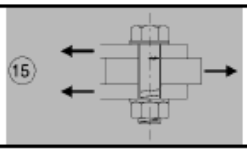
This section focuses on the fatigue calculation of a single bolt in the assembly shown in Figure 3. First the stress range using equation (6) will be obtained, followed by the calculation of forces and finally the torque amplitude.

3.3.1 Stress range

Using information provided by the company, all the variables for equation (6) were calculated.

- k_s = 1 since the bolt cross-section is less than 25mm.
- FAT class was obtained from Table 8.1 in EN 1993-1-9. FAT = 100 MPa. The table also provides m = 5.

Table 8.1 (continued): Plain members and mechanically fastened joints

Detail category	Constructional detail	Description	Requirements
100 m=5		Bolts in single or double shear Thread not in the shear plane 15) - Fitted bolts - normal bolts without load reversal (bolts of grade 5.6, 8.8 or 10.9)	15) $\Delta\tau$ calculated on the shank area of the bolt.

- The partial safety factor, γ_{Mf}, was obtained from Table 3.1 in Section 4 of EN 1993-1-9. In this case, a conservative approach is utilized so the safe life assessment method is appropriate. However, as described by information provided by the company, the bolts and pins connection is designed with a high factor of safety, so failure of a single bolt would not result in failure of the entire equipment, thus the consequence of failure is low. γ_{Mf} = 1.15 is used in this scenario.

Table 3.1: Recommended values for partial factors for fatigue strength

Assessment method	Consequence of failure	
	Low consequence	High consequence
Damage tolerant	1,00	1,15
Safe life	1,15	1,35

- The number of cycles, N_R is estimated as:
  - Total operation time (Months) = 14

- Total operation time (Days) = 14 × 30 = 420
- Total operation time (Minutes) = 420 × 24 × 60 = 604,800
- Typical RPM during operation = 750
- Total number of cycles, N_R = 750 × 604,800 = 4.54 × 10^8

Substituting the variables into equation (6):

$$\Delta\sigma_R = \frac{1 \times\ 100\ \text{MPa}}{1.15} \times \left(\frac{2 \times 10^6}{4.54\ \times\ 10^8}\right)^{1/5} = 29.39\ \text{MPa}$$

3.3.2 Force range

Using company reference information, the torque applied by the rotor is 4,244 Nm during normal operations (including a safety factor of 1.5). The distance of a single bolt from the centre of the shaft, *r*, is approximately 0.1m. The force on a single bolt can be calculated as:

- Torque = 4,244 Nm
- Torque acting on single bolt = $\frac{4244\ \text{Nm}}{5}$ = 848.8 Nm
- Torque acting on single bolt (excluding safety factor) = $\frac{848.8\ Nm}{1.5}$ = 566 Nm
- Force acting on single bolt = $\frac{566\ \text{Nm}}{0.1\ \text{m}}$ = 5,660 N

This force can be assumed to represent the minimum force (F_{min}) acting on the bolt under cyclic loading, as it is the force that is experienced during normal operation. Cyclic loading arises from power fluctuations in the motor causing the force acting on the bolt to vary from minimum to maximum values. Using equation (7), the maximum force (F_{max}) acting on the bolt can be obtained.

- Cross-sectional area of bolt = $\pi\ \times 5\ \text{mm}^2 = 78.54\ \text{mm}^2$

Plugging the values into equation (7):

$$29.39\ MPa = \frac{F_{max} -\ 5{,}660\ \text{N}}{78.54\ \text{mm}^2};\ F_{max} =\ 7{,}968\ \text{N}$$

Using equation (8), the maximum torque can be calculated:

$$0.1\ \text{m} \times 7{,}968\ \text{N} = 796.8\ \text{Nm}$$

Since the bolts are experiencing single shear stress, the minimum and maximum stresses on each bolt can also be obtained:

- Minimum shear stress = $\frac{5{,}660\ N}{78.54\ \text{mm}^2}$ =72.07 MPa
- Maximum shear stress = $\frac{7{,}968\ N}{78.54\ \text{mm}^2}$ = 101.46 MPa

4 Discussion

This section delves into a critical analysis of the results for each equipment type. The main aim is to provide an answer to the research questions posed at the beginning of the thesis based on the results obtained.

4.1 Pressurized vessel

The deformation results from the physical test generally followed the results predicted by the static simulation, supporting the reliability of the simulation model. As the simulation is based on a linear static analysis, the deformation exhibits a linear relationship with applied pressure, which is not fully representative of the nonlinear behaviour of the equipment in the physical test, as expected. However, at pressure levels closer to maximum values, the absolute differences between the simulated and physical deformation values are limited to nearly $1/10^{th}$ of a millimeter, which is considered small. The largest deviation occurs at the lower gauge location near the support legs, approximately 0.11mm. The full numerical results, including absolute and percentage differences, are provided in Appendix 1.

The observed differences may be attributed to several factors, including the assumption of perfectly linear elastic material behaviour in the static simulation, which does not fully capture potential material imperfections in real conditions. In addition, weld geometries were not modelled, which may influence the structural response of the vessel. Measurement errors during the physical test may also contribute to the differences. Despite these deviations, the model can be used as a sufficiently accurate approximation of the real vessel, and the results can be considered reliable for analysis purposes.

The wall thicknesses calculated using the EN 13445 standard and Maximum Principal Stress theory resulted in the same value of 2.35mm. This reflects the fact that the circumferential stress is the governing stress component to calculate the wall thickness of a pressure vessel. However, a thickness of 2.35mm is clearly insufficient for industrial process equipment. As per the note in clause 7.4.1, the thickness obtained from the standard represents a minimum value, and additional allowances must be included to account for non-pressure-related effects such as corrosion. Furthermore, clause 5.2.2 states that if it is expected that the wall

thickness will be reduced as a result of corrosion, then a sufficient increase in the thickness needs to be provided to ensure the required design life is achieved. The standard does not provide a specific value for this increase, leaving it to the engineer's judgement.

The thickness in the design drawings is 12mm, approximately 10mm higher than the calculated thickness. This suggests that an allowance for wear/erosion has been made in the design of the equipment, but it is possible that a 10mm difference may indicate excessive material usage.

Based on the results of the static simulation using the reduced wall thicknesses, it is clear that 4mm is insufficient, as the allowable stress (based on the safety factor limit) is exceeded in certain regions of the vessel.

In the case of the 6 mm wall thickness, the results indicate that this thickness value is generally acceptable, as the maximum stress level is approximately 111 MPa, which remains below the allowable limit. The small regions shown in red correspond to localised stress concentrations associated with abrupt geometric transitions, rather than a global structural issue. Such stress hot spots are common in areas with sharp changes in geometry and are not considered critical when they are highly localised. If these high stress regions were more extensive over large areas, they would represent a more significant design concern; however, this is not observed in this case.

Nevertheless, these results do not account for corrosion allowance. Therefore, a wall thickness of 8 mm could be considered a more reasonable design choice. This would reduce the total equipment mass to approximately 1.2 tonnes, corresponding to a material saving of 23% compared to the original design. However, further investigation into wear/erosion effects would be required to confirm whether this thickness is suitable for the intended design life of the equipment.

4.2 Large Non-Pressurized Mixing Vat

Based on the calculated conical wall thickness value of 1.23mm and the results of the static simulation, it is clear that the mixing vat has a safe design. The wall thickness value of 8mm results in stress levels well below the safe limit, indicating excessive material usage and potential for structural optimization.

As stated previously, the analytical value only considers the hydrostatic pressure in the calculation and ignores the effects of other loading types such as the rotor mass, cyclic loading and wear. Given the minimum value of 1.23mm, it would be reasonable to add a margin of a several millimetres, resulting in 5mm as a proposed thickness value.

Similarly with the static simulation, as it also ignores the effect of wear or cyclic loading, the results should be interpreted with these limitations. Even so, based on the low stress levels experienced using 5mm walls, this might be a reasonable choice for the conical wall thicknesses.

In addition to a material saving of 7%, an added advantage of using 5mm for the conical sections is that the entire main body of the vat (cylinder plus bottom cone) can be constructed with a single metal sheet of the same thickness. This would reduce material waste at the fabrication stage and result in a more simplified manufacturing process.

As a side note, the stress in some regions of the big baffle exceeds the yield strength of the material when the support plates behind the baffle are not included in the simulation. Once the support plates are added in, the average stress drops significantly to 21 MPa from 67 MPa on the body of the baffle, with no areas exceeding the yield strength. However, the baffle assembly is outside the scope of this study, as it is already using a 6mm wall thickness, and so any changes would not lead to a meaningful reduction in material usage.

The symmetrical study demonstrated that the inclusion of the cover results in a much stiffer structure and would therefore be useful for a vat of this size. The average stress was largely unchanged, as the stress levels using the 8mm thickness value were already quite low.

4.3 Non-Pressurized Mixing Vessel

The material used for the bolts is A2-70. This is a class of stainless steel that is commonly used for fasteners due to its excellent corrosion and durability properties. It is designed to last in demanding industrial applications such as the equipment analyzed in this thesis, with a minimum yield strength of 205 MPa (SteelPro Group, 2024).

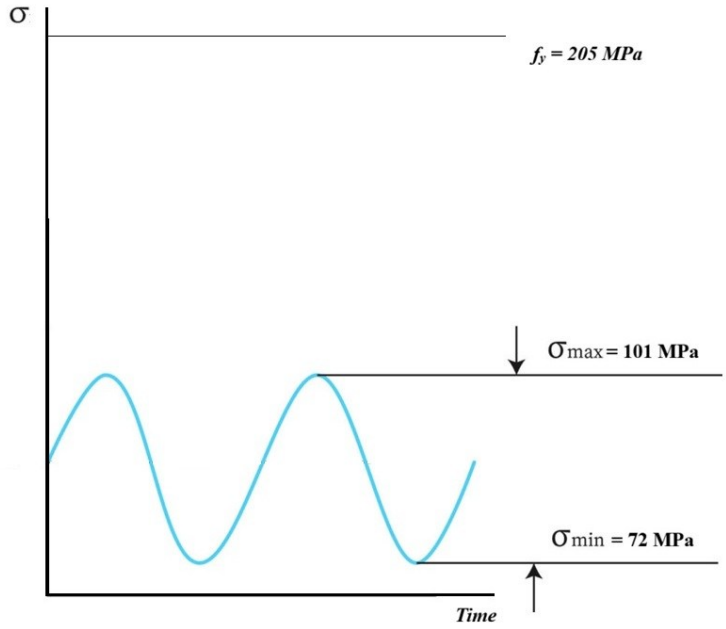


Figure 26 - Fatigue loading

Based on the results obtained in section 3.3.2, the maximum shear stress experienced by a single bolt is 101 MPa, while the minimum (which represents normal operating load) is only 72 MPa. As depicted in Figure 26, this cyclic loading is way below the minimum yield strength of the bolt material. This is one of the characteristics that makes fatigue a dangerous failure mechanism, as the loading seems to be within safe limits, but is actually leading to cracks that are very difficult to detect and sudden failure of the component.

This result using the nominal stress method shows the nature of the loading experienced by each bolt given a finite fatigue life. The result shows that in normal operating conditions, the loading is insufficient to cause the bolt to break due to yielding, so the most likely reason for bolt failure would be cyclic loading within the range shown in Figure 26. While the nominal stress method used provides a useful initial approximation, the actual fatigue behaviour may differ due to several factors that are not considered in the analysis:

- Stress concentrations and notch stresses, which are not used in this method.
- The real system would not experience a constant amplitude loading as assumed in the calculation. There would be variability due to vibrations and small fluctuations in the power delivered by the motor.
- Bolt preload, which would introduce friction between the clamped surfaces, is not taken into account. This would impact the stress experienced the bolts.

- Wear and corrosion effects are not included in the fatigue calculation.

A more detailed analysis using other fatigue assessment methods would need to be performed which includes the factors above. However, these can be resource-intensive and their use would need to be justified. For the purpose of obtaining an initial estimate, the nominal stress method used in this study is considered appropriate.

5 Conclusion

The main objective of this thesis was to evaluate whether the selected process industry equipment uses designs that are overly conservative. Using FEA, the project aimed to identify the major opportunities where material could be optimized in the existing design while maintaining an acceptable safety margin. This was successfully demonstrated by the study with material savings of 23% and 7% for the pressurized vessel and mixing vat respectively. These results were supported by analytical calculations used to calculate minimum wall thicknesses for each piece of equipment being analysed.

In addition to the static analyses, a fatigue study was a performed as a supplement to the overall project to analyze the mechanical bolted connection of separate type of process equipment. Using the nominal stress method, the cyclic loading and associated stress on each bolt used in the connection were calculated, which showed loading levels way below the yield strength of the bolt material, highlighting the critical nature of fatigue failure. The study reinforces the need for consideration of fatigue damage in industrial equipment design.

While the static studies performed in this thesis have limitations, the results provide a compelling reason for equipment manufacturers to re-think current design approaches. Optimization of material usage would benefit company profitability and market competitiveness and simultaneously contribute to broader positive effects for the environment. Future work to extend this research could integrate dynamic analyses that include transient loading and the effects of wear/corrosion. Combining these studies with the static simulation results presented here would provide a solid foundation for optimization of equipment designs. In the long-term, the benefits of transitioning towards more material efficient designs are expected to outweigh any short term-costs, supporting more cost-effective and sustainable design practices in the process industry.

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Appendices

Appendix 1 - Pressurized vessel Deformation Result Comparison

	Ansys Deformation (mm)			Physical Test Deformation (mm)			Absolute Difference (mm)			Percentage Difference		
Pressure (MPa)	Front	Middle	Low	Front	Middle	Low	Front	Middle	Low	Front	Middle	Low
0.049467	0.06	0.03	0.02	0.00	0.00	0.00	0.06	0.03	0.02	100%	100%	100%
0.098934	0.12	0.06	0.04	0.04	0.05	0.01	0.08	0.01	0.03	68%	16%	76%
0.148401	0.18	0.09	0.06	0.04	0.05	0.01	0.14	0.04	0.05	78%	44%	84%
0.197868	0.25	0.12	0.08	0.15	0.10	0.04	0.10	0.02	0.04	39%	15%	51%
0.247335	0.31	0.15	0.10	0.21	0.17	0.04	0.10	-0.02	0.06	31%	-16%	61%
0.296802	0.37	0.18	0.12	0.30	0.19	0.06	0.07	-0.01	0.06	18%	-8%	51%
0.346269	0.43	0.21	0.14	0.36	0.23	0.07	0.07	-0.02	0.07	16%	-12%	51%
0.395736	0.49	0.23	0.16	0.36	0.23	0.07	0.13	0.00	0.09	26%	2%	57%
0.445203	0.55	0.26	0.18	0.45	0.23	0.07	0.10	0.03	0.11	18%	13%	62%
0.49467	0.61	0.29	0.20	0.53	0.26	0.10	0.08	0.03	0.10	13%	11%	51%